Problem 9

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Problem 1 *Suppose the function f has derivative given by $f'(x) = \dfrac{2x}{4+x^2}$.*

(a) *Find a formula for $f''(x)$.*

Explanation. The second derivative is given by $f''(x) = \dfrac{2(4-x^2)}{(4+x^2)^2}$.

(b) *Construct a sign-chart for f' and f''. [2]*

Explanation. For f' we obtain:

	$(-\infty,\ 0)$	$(0,\ \infty)$
$2x$	$-$	$+$
$4+x^2$	$+$	$+$
$f'(x) = 2x(4+x^2)$	$-$	$+$

For f'', using the factorization $4-x^2 = (2+x)(2-x)$, we obtain:

	$(-\infty,\ -2)$	$(-2,\ 2)$	$(2,\ \infty)$
2	$+$	$+$	$+$
$2+x$	$-$	$+$	$+$
$2-x$	$+$	$+$	$-$
$(4+x^2)^2$	$+$	$+$	$+$
$f''(x)$	$-$	$+$	$-$

(c) *On what interval(s) is f increasing? On what interval(s) is f concave down?*

Explanation. From the sign-chart for f', we see that f' is positive on $(0,\infty)$ and negative on $(-\infty,0)$. We know that f is increasing on an interval if f' is positive on that interval. That means: f is increasing on the interval $(0,\infty)$.

From the sign-chart for f'', we see that f' is positive on $(-2,2)$ and negative on $(-\infty,-2)$ and $(2,\infty)$. We know that f is concave down on an interval if f'' is negative on that interval. That means: f is concave down on the intervals $(-\infty,-2)$ and $(2,\infty)$.